

Student Name:

Teacher:

**Island School Science
Year 9
Our Planet
Student Booklet
(Revised June 2025)**



Yr9 1a The Impact of Acid Rain on the Environment

Introduction:

Acid rain is a complex environmental problem that concerns many environmental and chemical engineers.

When engineers examine the acid rain damage to water, wildlife, forests, crops and structures, they consider the impact on human health. Engineers design many useful technologies that help industry reduce the amount of harmful pollutants released into our air.

Engineers also help to develop laws that prevent or limit factories and industries from burning fossil fuels (which release pollutants), or require them to minimise their pollutant output.

Aim:

To investigate the negative impact of acid rain on living and non-living matter.

Method:

1. Using the universal indicator and/or a pH probe measure the pH of the vinegar and the distilled water, and record it on their worksheets.
2. You must now use your scientific knowledge to date to make some predictions. If vinegar contains acid (ethanoic acid), then how will the items placed in vinegar change? If these items are placed in water, will they change in the same ways as in the vinegar?
3. Use masking tape and pens to label one jar/beaker "vinegar" and the other one "water."
4. Using an electric balance measure the mass of the leaf, paper clip and egg shell and record in the results table below.
5. Pour 100cm³ vinegar into the vinegar jar. Place a paperclip, piece of eggshell and a green leaf in the vinegar. Put the lid on the container.
6. Pour 100cm³ of distilled water into the water jar. Place a paperclip, piece of eggshell and a green leaf in the distilled water. Put the lid on the container.
7. Let the jars sit overnight on a windowsill or protected area.
8. The next day, remove the container lids. Observe any changes in the condition of the items in the jars. [You may need to come to the lab during a convenient time for your teacher to make these observations] Enter your initial observations in the results table provided.
9. **After one week**, look for more changes. Make observations again. Remove the paper clip, egg shell and leaf and measure the DRY mass of each. Record your mass readings in your results table.

Results and Analysis

Record in the table below your predictions (what you think will happen to each item). Next, record your observations (a detailed description of what *actually* happened to each item) after one day in each solution and after one week in each solution.

	Distilled Water [pH =]			Vinegar [pH =]		
	Predictions	Observations		Predictions	Observations	
		After 1 Day	After 1 Week		After 1 Day	After 1 Week
Eggshell						
Green Leaf						
Paperclip						

Acid:

Object	Mass in Day 1 (g)	Mass in Day 7 (g)	Change in Mass (g)
<i>Leaf</i>	2.5	2.1	-0.4
<i>Eggshell</i>	1.8	1.2	-0.6
<i>Paperclip</i>	0.5	0.5	0.0

Water:

Object	Mass in Day 1 (g)	Mass in Day 7 (g)	Change in Mass (g)
<i>Leaf</i>	2.4	2.3	-0.1
<i>Eggshell</i>	1.7	1.7	0.0
<i>Paperclip</i>	0.5	0.5	0.0

Conclusion:

1. Describe what happened to the living items (the leaf and eggshell) tested in the acidic solution.

The leaf became soft, limp and darker as the acid dissolved parts of it, and the eggshell became soft and started to dissolve and fizz, because the acid reacted with the calcium carbonate in it.

2. Do you think acid rain has this effect on all living things? Explain why or why not.

No. Only living things with hard calcium carbonate parts or thin soft tissue are broken down quickly; some plants and animals tolerate acid, but acid rain weakens most living things over time.

3. Describe what happened to the non-living item (paperclip) tested in the acidic solution.

Very little happened - the paperclip showed no visible change and stayed shiny, because the metal does not react with the weak acid in the vinegar.

4. Do you think acid rain has this effect on all non- living things? Explain why or why not.

No. Some non-living things do react with acid - limestone, marble, concrete and metal structures corrode and are eaten away by acid rain - so it depends on the material.

5. Did any of the object's change in mass? Explain any changes.

Yes. The eggshell and leaf both lost mass because the acid reacted with and dissolved them, while the paperclip lost no mass because it did not react; the items in water barely changed.

6. Why is acid rain harmful to forests?

Acid rain damages leaves and needles so the trees make less food, and it leaches nutrients from the soil and releases toxic aluminium, so the trees weaken and die from disease, pests or cold.

Acid Deposition

What Is Acid Deposition?

Acid deposition includes acid rain, snow, fog, humidity and dust with an acidity level lower than pH 5.6. Normal rain, which has a pH of about 5.6, is about 25 times more acidic than “pure,” neutral water. The acidity results from the conversion of atmospheric carbon dioxide in water vapor to carbonic acid, a weak acid. Most acid precipitation has a pH between 4.6 and 5.6, but scientists occasionally have measured pH values in acid rain in the eastern U.S. as low as 2.1 and 3.0, which is about 10,000 to 80,000 times more acidic than pure water.

From Where Does It Come?

Nearly 95% of the acidity below pH 5.6 comes from atmospheric sulfur dioxide and nitrogen oxides, which are products of fossil fuel combustion. Acidified rainwater contains combinations of sulfuric and nitric acids that form when water vapor, and sulfur dioxide and nitrogen oxides react. Most of the acid deposition in the eastern U.S. is attributed to the release of large amounts of sulfur dioxide and, to a lesser extent, nitrogen oxides from big mid-western power plants that burn coal. Paper and wood pulp processing plants also contribute to sulfur dioxide pollution. In the U.S. and Canada, sulfur dioxide contributes much more to acid deposition than nitrogen oxides, which come mainly from automotive emissions, but over the next few decades, nitrogen oxides may catch up. Sulfur and nitrogen oxides may be transported by the wind in the atmosphere for many miles, crossing regional and international boundaries, before falling to Earth.

What Are Its Effects?

The effects of acid rain may not be immediately apparent. For example, at a glance, a lake might look clear and beautiful, but a closer look may reveal few living organisms. Some species of fish cannot

reproduce in water with a pH of less than 5. Clams, snails, crayfish and other crustaceans, brook trout, walleyed pike and bullfrogs are especially sensitive to acidification. However, the detrimental impact of acidification to animal life is not necessarily caused directly by the acidity. Trace metals such as aluminum, mercury, manganese and cadmium, which are leached from sediment and rocks by the increased acidity, are toxic to life. Thus, the pH does not have to decrease very much before fish kills can occur.

Because many insects cannot survive in strongly acidic streams and lakes, birds and mammals that depend on insects for food may suffer abnormally high mortality.

Acidification also interrupts normal decomposition of dead plant and animal material in lakes and streams because many of the bacteria that assist in decomposition perish. Without the usual decomposition processes, dead material settles to the bottom, making the water look crystal clear.

The damaging effects of acid deposition on forests and other terrestrial systems are less well understood than on aquatic systems. Acid deposition can alter soil chemistry, nutrient availability and plant growth. In their weakened condition, trees and shrubs become vulnerable to insects, diseases and fungus infestations.

Although the Norwegians were the first to bring acid rain to the world’s attention in the 1940s, one of the most severely impacted areas of the world is the industrialized Ruhr Valley in West Germany. There, white fir trees became defoliated and died in the early 1970s. Diseases in spruce and other sensitive conifers soon followed, and by 1985, the number of German trees visibly affected by acid deposition had risen to 52%. Forests in other parts of the world also display acid deposition damage. For example, the

dominant tree in Vermont's Green Mountains, the red spruce, is suffering severe mortality and parts of the mountain range have become denuded. Sugar maples all over the northeastern U.S. and Canada are declining. In the Shenandoah and Great Smokey Mountains of the southeastern U.S., spruce and fir are failing to reproduce and are dying. Pine also are impacted.

How Do We Recognize It?

Particulate matter containing atmospheric sulfur dioxide and nitrogen oxide account for over 50% of the visibility problems in the eastern U.S. In the West, these particles have been blamed for reducing visibility in the Grand Canyon of the Colorado River and other areas. Acid deposition contributes to the corrosion of metals and the deterioration and soiling of the stone and paint on buildings, statues, and other structures of cultural significance.

How Do We Reduce Its Effects?

The U.S. federal government has undertaken a wide range of research programs, many through the National Acid Precipitation Assessment Program, to study the complex processes associated with acid rain. To measure acid deposition quantity and chemistry, scientists collect rainfall samples at monitoring stations throughout the U.S. Monitoring dry deposition such as acid dust is difficult and, consequently, has not been as extensive as that for wet deposition.

The Clean Air Act Amendments of 1990 established an Acid Rain Program to reduce emissions of sulfur dioxide and nitrogen oxides at the lowest cost to society. To

achieve reductions of 10 million tons of sulfur dioxide by the year 2010, the Act requires a two-phase tightening of the restrictions placed on fossil-fuel-fired power plants. Phase I began in 1995 and affects 110 coal burning electric utilities in 21 mid-western and eastern states. Phase II began in 2000 and tightens annual emissions on the large plants and also sets restrictions on smaller, cleaner plants burning coal, oil and gas. To achieve reductions of 2 million tons of nitrogen oxides by 2000, the Act requires coal-fired utilities to install low-nitrogen-oxide technologies on their burners.

To reduce sulfur dioxide emissions, fossil fuel burning plants can burn low-sulfur coal, install flue-gas desulphurization equipment (scrubbers) and implement clean combustion technologies. Low-sulfur coal contains about 1% sulfur by weight and is found mainly in the western U.S. High-sulfur coal contains sulfur in excess of 3% and is geographically concentrated in the Appalachians where coal utilization is greatest. Scrubbers are effective at reducing air pollution, but the sludge they produce creates a disposal problem. Also, they are expensive to build and operate. Clean combustion technologies involve mixing fuels with compounds that react with sulfur and either collect it or convert it to a marketable product that does not enter the stack. Many of these innovative technologies have the added advantage of removing nitrogen oxides as well. Because emissions from motor vehicles are a major source of nitrogen oxides, catalytic converters are used to reduce nitrogen oxides from automotive sources.

Source: Project A.I.R.E., U.S. Environmental Protection Agency,
<http://www.epa.gov/region01/students/teacher/aire.html>.

9 OP L2a

Causes/Reasons for deforestation	Effects/consequences of deforestation
Clearing land for farming and cattle ranching; building homes, roads and factories; logging for timber, paper and furniture; collecting firewood; mining; and natural forest fires.	Habitat destruction - animals lose their homes and may become endangered or extinct; soil erosion, murky rivers and landslides; more carbon dioxide in the air, adding to climate change.

Deforestation in the Amazon Rainforest

Introduction

Deforestation is the removal or clearing of a large area of trees. Deforestation may happen for several different reasons. One reason is to build homes, buildings, and factories for people to live and work in. Another reason is to make space for farming and agricultural activities. Sometimes people remove trees to make items from them, such as paper or furniture. Trees may be cut down to use as firewood. Many times deforestation is human-caused, but trees can also be cleared after a natural event, such as a tornado or fire.

When people remove trees from an area, they can use different techniques. One is called **clear-cutting**. People will cut every single tree in an area. Another way people may remove trees is called **slash-and-burn**. In the slash-and-burn technique, the trees are cut down and then burned. Sometimes this is done so that the ashes from the burned trees go into the soil, which provide nutrients for farm crops to grow in place of the trees.



There are several effects of deforestation. Deforestation is a type of **habitat destruction**, and can cause many animals to lose their homes. They have to find another place to live, or they will not survive. Some organisms may become **endangered** or even face **extinction**, if an area is completely destroyed. Deforestation can also lead to **erosion**. The roots of the trees help soil stay in place, and when the roots are gone, the soil becomes loose. It can move around, being carried by wind or water, causing murky rivers and lakes, or even landslides. Lastly, deforestation can lead to an increase in **greenhouse gases**. Trees take in greenhouse gases like carbon dioxide when they do **photosynthesis**. When the trees are gone, more greenhouse gases build up in the atmosphere and can contribute to **climate change**.

In this activity, we will investigate how the **Amazon Rainforest** has changed in size due to deforestation in the past 50 years. We will create a line graph of the changes over time of the amount of forested area.

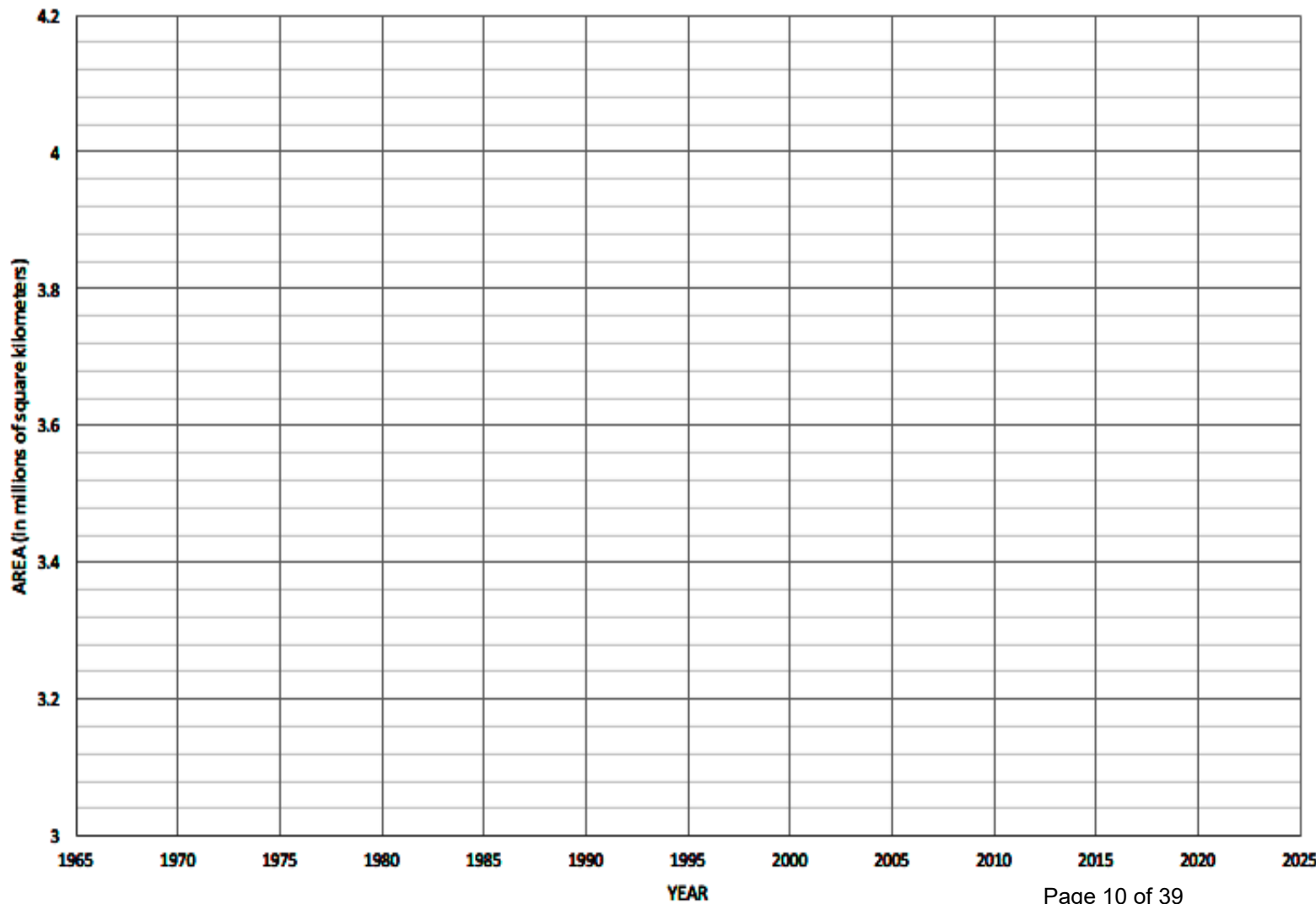
Directions

The chart on the right shows the size of the Amazon rainforest, in millions of square kilometers. This means that in 1970, 4.1 million km², or 4,100,000 of rainforest area existed in the Amazon. The data is from Brazil's Institute for Space Research (INPE).

Plot a point on the graph for each year in the chart below. Then, using a straight edge, connect the dots from left to right to create a line graph.

Deforestation in the Amazon	
Year	Size of Rainforest (in millions of km ²)
1970	4.10
1977	3.96
1985	3.86
1990	3.80
1995	3.71
2000	3.60
2005	3.50
2010	3.43
2015	3.42
2019	3.38

Area of Rainforest in the Amazon



Analysis and Conclusion

1. Describe the pattern you see in the graph.

The graph shows a steady decrease - the rainforest shrank from 4.10 million km² in 1970 to 3.38 million km² in 2019, a loss of about 0.72 million km², with the steepest fall between 1970 and 1990.

2. What years does the graph decrease the slowest? Why do you think this happened?

The decrease is slowest between 2010 and 2019 (3.43 to 3.38 million km²), because of tougher laws and better monitoring in Brazil, protected reserves and pressure from conservation groups.

3. Predict what you think will happen to the area of Amazon rainforest in the future. Will it increase, decrease, or stay the same? Justify your answer with reasoning.

It will probably keep decreasing, but more slowly, because demand for farmland and timber continues; unless forest protection laws are enforced and land is replanted, the area will shrink further.

4. In 1970, there were an estimated 120,000 jaguars in South and Central America. In 2014, there were estimated to be about 12,000 jaguars. Explain how this change in the jaguar population may have been affected by deforestation.

Deforestation destroyed the jaguars' habitat and reduced the prey they hunt, so fewer could survive and breed - the population fell from 120,000 in 1970 to about 12,000 in 2014.

5. How can people help make a difference and **conserve** the rainforest, even when we don't live near it? Give at least two ideas.

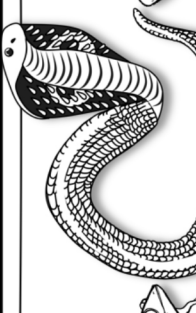
Buy certified sustainable products (coffee, chocolate, wood, palm oil), eat less beef, recycle paper, support rainforest charities and reforestation, and raise awareness of deforestation.



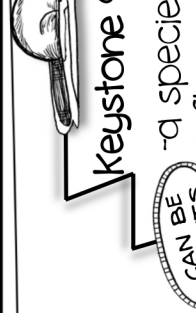
A group of similar organisms that have similar characteristics to each other and can interact with each other.



Biodiversity











Economic Value

-many organisms provide people with _____ and supply raw _____ for clothing, medicine, and other products

-ecosystems provide _____ for people



Ecological Value

-all species in an ecosystem are _____ to one another

-species _____ on one another for food and shelter

-a change that affects one can affect _____



Threatened Species

-species that could become _____ soon



Endangered Species

-species in danger of becoming _____ soon

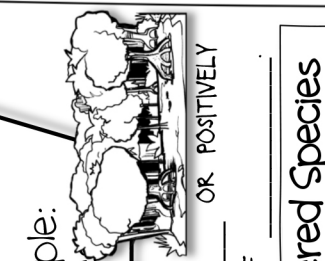
SEA OTTERS



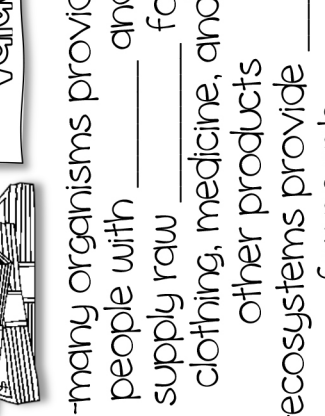
Keystone Species

-a species that influences the _____ of many other species in an ecosystem

Example: _____

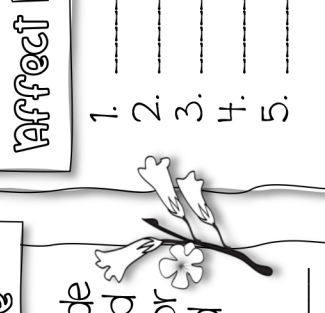


NO ONE KNOWS HOW MANY SPECIES HAVE NOT YET BEEN



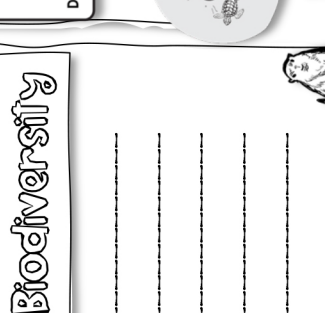
Factors that Affect Biodiversity

- _____
- _____
- _____
- _____
- _____



Plant & Animal Species Diversity on Earth

There are many more species of _____ than of _____ or other _____ species.



Wild Organisms are a Source of Food and Recreation.



Biodiversity from Place to Place on Earth.



Ecological Value

-all species in an ecosystem are _____ to one another

-species _____ on one another for food and shelter

-a change that affects one can affect _____



Threatened Species

-species that could become _____ soon



Endangered Species

-species in danger of becoming _____ soon

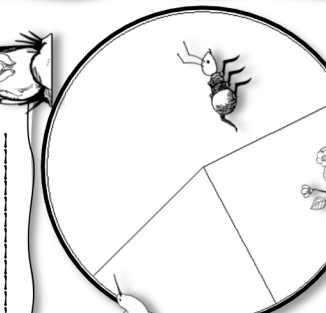
SEA OTTERS



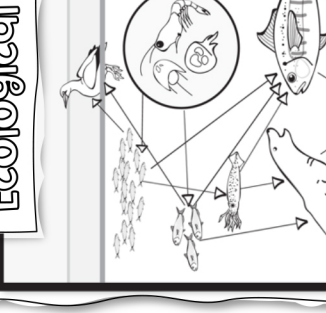
Keystone Species

-a species that influences the _____ of many other species in an ecosystem

Example: _____

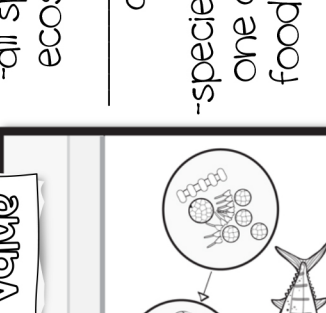


NO ONE KNOWS HOW MANY SPECIES HAVE NOT YET BEEN



Factors that Affect Biodiversity

- _____
- _____
- _____
- _____
- _____



Plant & Animal Species Diversity on Earth

There are many more species of _____ than of _____ or other _____ species.



Wild Organisms are a Source of Food and Recreation.



Biodiversity from Place to Place on Earth.



Ecological Value

-all species in an ecosystem are _____ to one another

-species _____ on one another for food and shelter

-a change that affects one can affect _____



Threatened Species

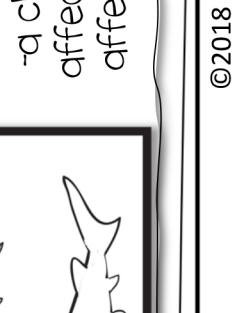
-species that could become _____ soon



Endangered Species

-species in danger of becoming _____ soon

SEA OTTERS



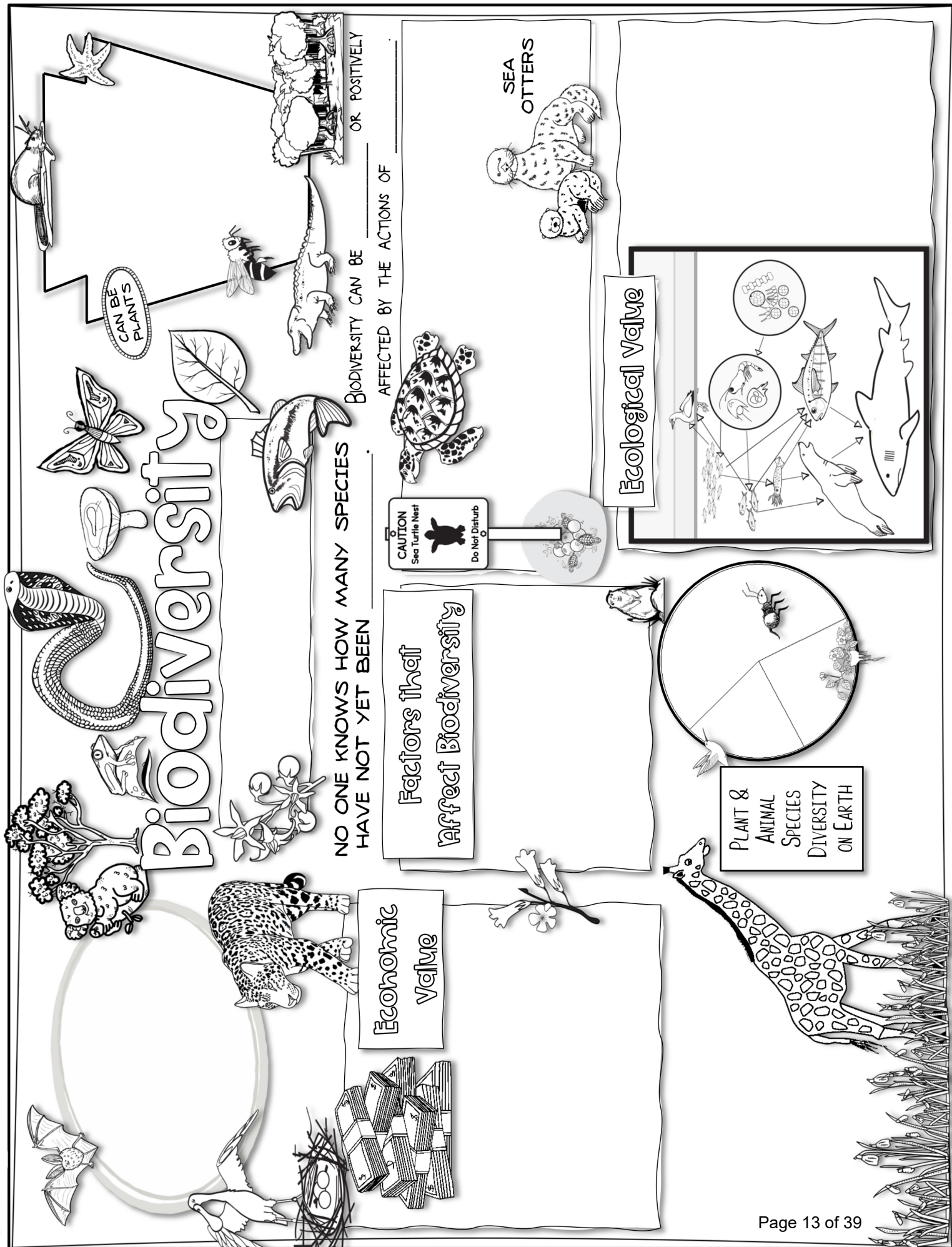
Keystone Species

-a species that influences the _____ of many other species in an ecosystem

Example: _____



NO ONE KNOWS HOW MANY SPECIES HAVE NOT YET BEEN



Biodiversity

CAN BE
PLANTS

NO ONE KNOWS HOW MANY SPECIES
HAVE NOT YET BEEN

BIODIVERSITY CAN BE
AFFECTED BY THE ACTIONS OF

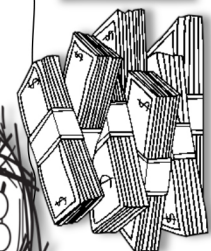
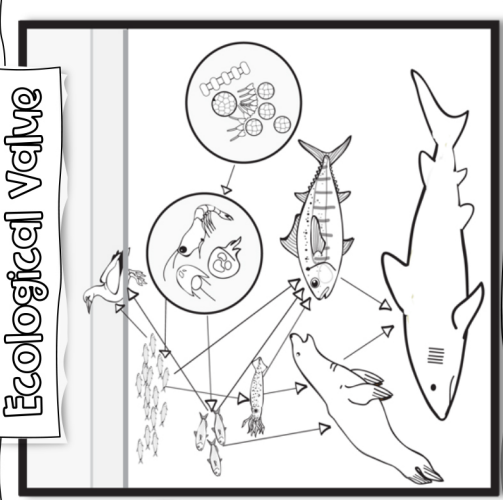
Economic
Value

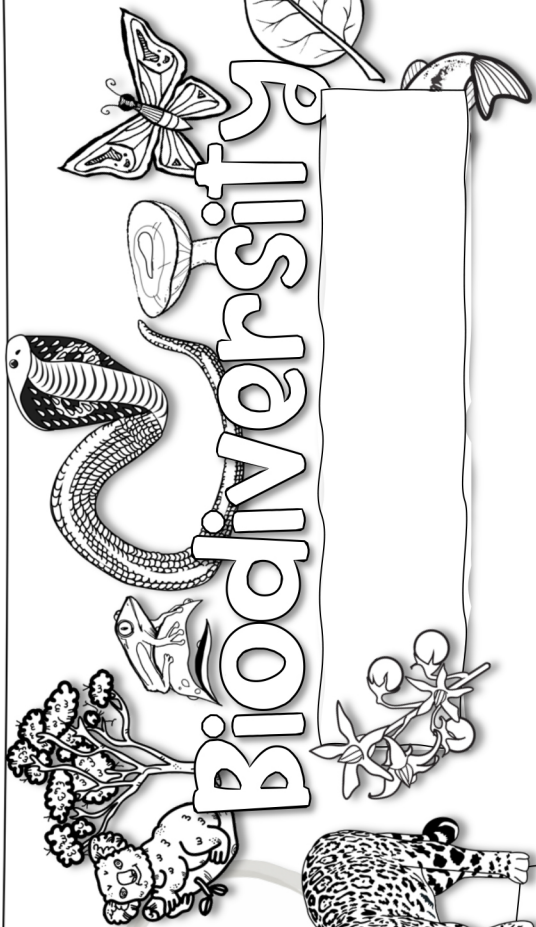
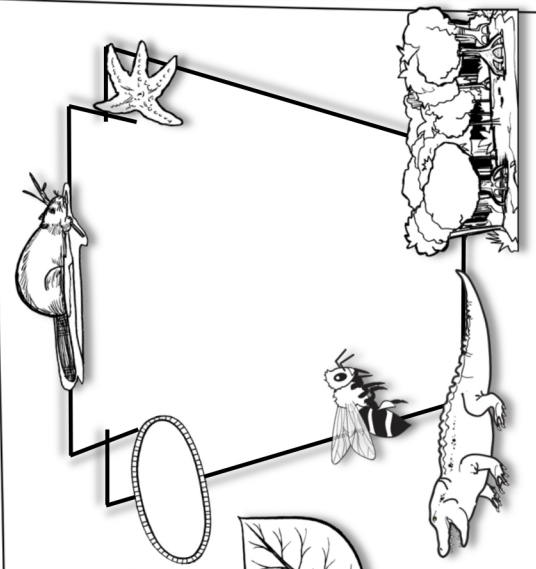
Factors that
Affect Biodiversity

Ecological Value

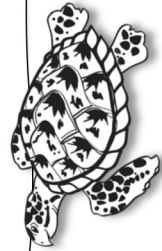
PLANT &
ANIMAL
SPECIES
DIVERSITY
ON EARTH

SEA
OTTERS

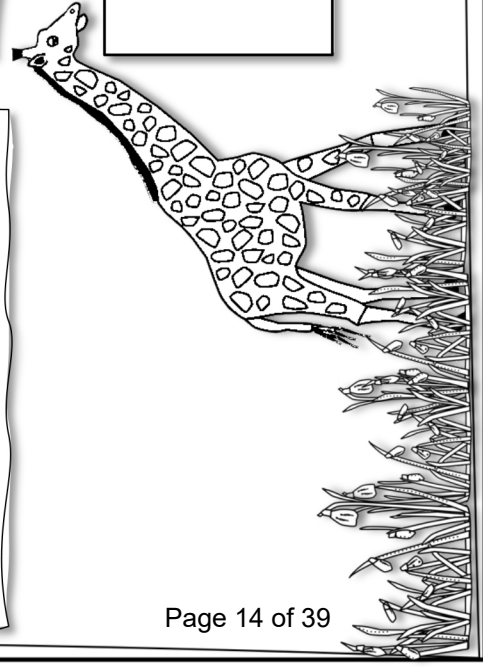
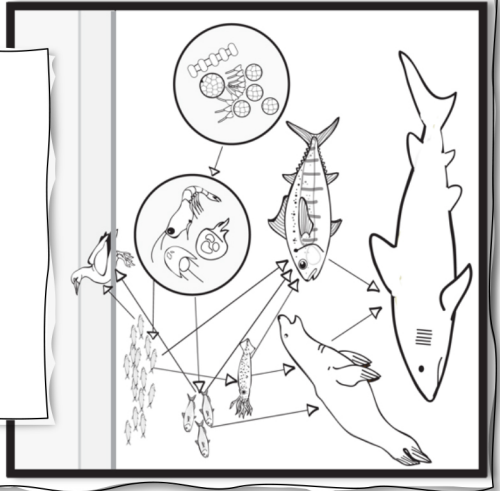
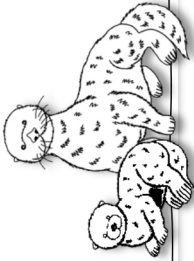




Biodiversity



CAUTION
Sea Turtle Nest
Do Not Disturb



Understanding Checkpoint



Biodiversity

Name _____ Date _____ Period _____

- Biodiversity is the variety of different _____ in an area.
 - The number of known species is about _____ million.
 - 17
 - 1.7
 - 10.7
 - 11.7
- A sea otter is critical to the functioning of the ecosystem in which it lives. It affects the survival and abundance of many other species in its community. It is a(n) _____ species.
 - keystone
 - endangered
 - threatened
 - unknown
3. A sea otter is critical to the functioning of the ecosystem in which it lives. It affects the survival and abundance of many other species in its community. It is a(n) _____ species.
 - keystone
 - endangered
 - threatened
 - unknown
4. Which organisms on Earth have the most species named?
 - plants
 - animals
 - bacteria
 - insects
5. Organisms have _____ value.
 - ecological
 - economic
 - recreational
 - all of these
6. When there are no individuals of a certain species left anywhere on Earth, they are considered
 - endangered.
 - extinct.
 - threatened.
 - keystones.
7. Organisms provide people with jobs and the things they need to live; that gives them _____ value.
 - economic
 - ecological
 - recreational
 - niche
8. This factor affects biodiversity:
 - climate
 - size of the area
 - genetic diversity
 - all of these
9. True or False: All species in an ecosystem are connected.
10. True or False: People have no influence on biodiversity.

What if the other species could talk back?

What if the species could talk? What if it is no longer a monologue but a dialogue? What if our dwindling species put us in a courtroom and ask- 'What are you doing?'

We have shown our incompetence and incapability in understanding their presence, plight, and plea. Today, you have a unique opportunity to express their viewpoint in the first person. You have the responsibility of representing them, and hopefully, you would give them a voice that will make a difference, even if just a little.

1. Work in groups of 3-4
2. Discuss the case history a critically endangered species
3. Research before you choose, look up species from different kingdoms (not just animals and not just mammals)
4. Explore the link below to find various species: <https://www.iucnredlist.org/>
5. You are a family representing the species of your choice
6. Prepare an **emotive speech** to show the **plight** of your species, **pressures** you are facing, the **ecological significant role** you play
7. You are presenting your case to a **jury** (teacher), the jury will choose **ONLY one species** that could be conserved as funding is an issue
8. **Why should your species be conserved?**
9. You will have **5 minutes** to present your case
10. The jury might **ask you specific questions** before making their decision so make sure that you have done good research

Include all of the following in your role play

- Name and description of the species
- Its ecological role
- Pressures (ecological, socio-political and economic) contributing to its IUCN threatened the status
- The species' ecological roles and the possible consequences of their disappearance should be considered
- You will need to know the ecological, political and economic pressures that caused or are causing the degradation of the chosen area, and the consequent threat to biodiversity

How will you present your research?

- You will bring **props**
- You will **dress up** as your species, be **creative** (dress code applies)
- Give a named ecosystem where your species comes from
- You will give a first-person account of what the species has faced or is currently facing
- You will be engaged in a maximum 5-minute long speech/ dialogue amongst your group members that cover the points mentioned above
- You will **submit the written speech** with **APA in-text** and works cited

Assessment Criteria:

- Research shown by your speech works cited and in-text referencing
- Creative thinking skills in the usage of props
- Critical Thinking skills

David Attenborough: A Life on Our Planet

1. What human disaster happened in the city of Chernobyl in 1986?

A nuclear disaster - the reactor at the Chernobyl nuclear power station exploded and caught fire, releasing radioactive material over the surrounding area.

2. How old is Sir David Attenborough?

94 years old (he was born in 1926 and the film was released in 2020).

3. What was the world population in 1937?

About 2.3 billion people.

4. What percentage (%) of species were wiped out when a meteor crashed into the Earth, killing all of the dinosaurs?

About 75% of all species were wiped out.

5. By 1954, what percentage (%) of wildlife was remaining?

64% of the world's wildlife was still remaining.

6. What does the word 'Serengeti' mean?

It means 'endless plains' - the Maasai name for the vast plains of East Africa.

7. In what year was the Apollo 8 mission launched?

December 1968.

8. Which ocean animals were being hunted in the 1970s?

Whales were being hunted.

9. By how much had Borneo's rainforest decreased by the end of the 20th century?

Borneo's rainforest had decreased by about half (50%).

10. Up to how many species of tree are there in a small patch of rainforest?

Up to 150 species of tree in a small patch of rainforest.

11. What colour did all of the coral turn?

White - the coral bleached.

12. What change happened in the ocean to cause coral to turn this colour?

The sea temperature rose (by around 1 degree C), so the coral expelled the tiny algae living inside it and turned white.

13. In the last 40 years, what percentage (%) of the summer sea ice in the Arctic has been reduced by?

By about 40%.

14. How many trees do we cut down each year?

About 15 billion trees are cut down each year.

15. What is the Earth's population in 2020?

About 7.8 billion people in 2020.

16. How many people are estimated to be on Earth by 2100?

About 11 billion people.

17. In which country is the world's largest solar farm?

India (the huge solar farm shown in the film, in Tamil Nadu).

18. How did people on the island of Palau respond when their fish stocks decreased?

They protected a large part of their ocean as a marine sanctuary, banning fishing there, so fish stocks recovered and the reef became healthier.

19. How could many humans change their diets to make it more sustainable?

Eat less meat, especially beef, and more plant-based food; buy local, seasonal food and waste less food.

20. Which greenhouse gas do trees absorb from our atmosphere?

Carbon dioxide.

21. What has happened to the city of Chernobyl since it was evacuated 30 years ago?

Nature has taken over - the forest has grown back and wildlife such as wolves, deer and birds has returned to the abandoned city.

22. Look at all of the words in the box below and circle/highlight all of the emotions which you felt whilst watching 'David Attenborough: A Life on Our Planet'.

· Circle all the emotion words in the box that describe how the film made you feel (e.g. sadness, shock, anger, fear, guilt, hope). The box is an image, so any emotions you felt may be chosen.

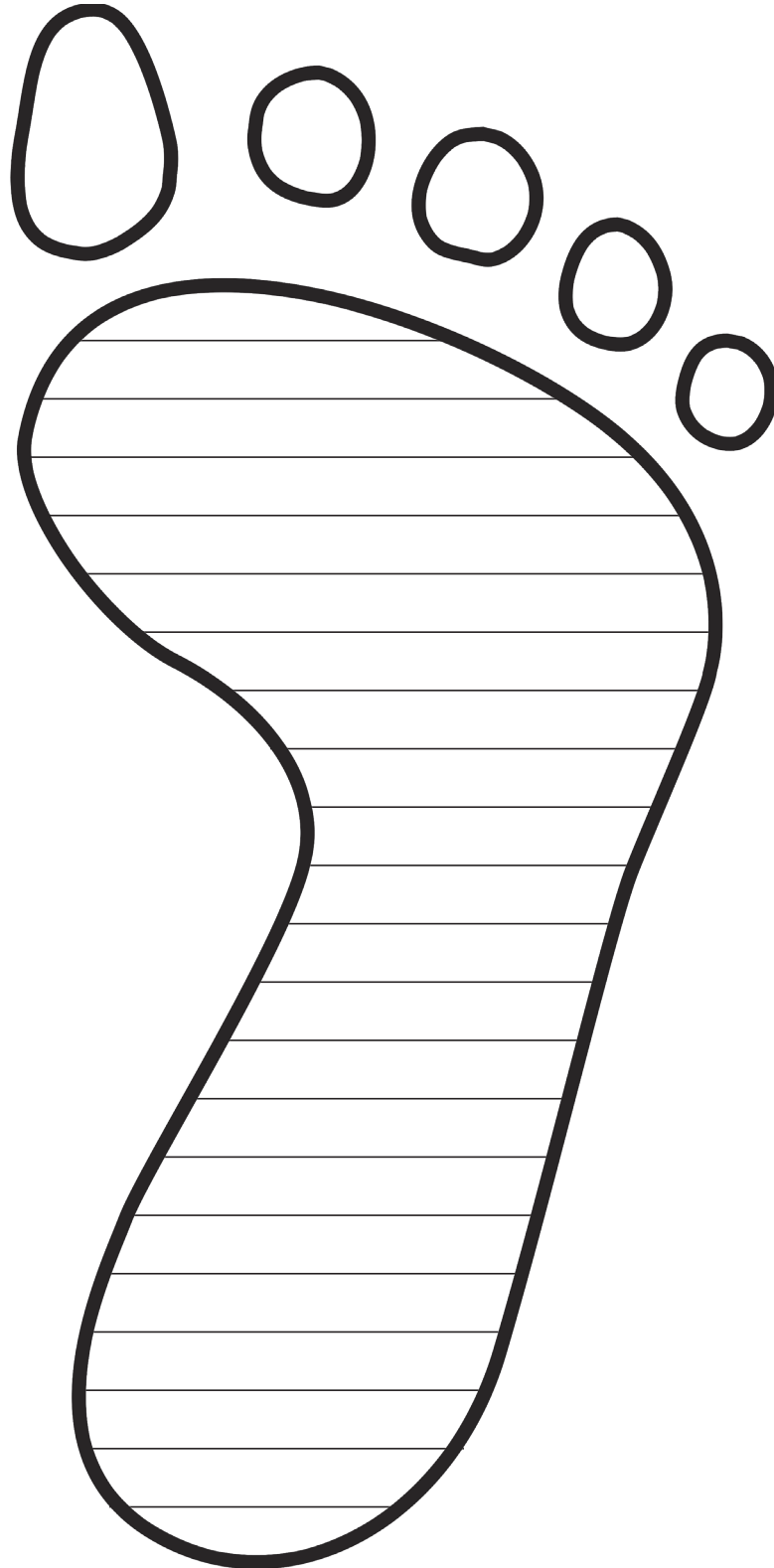
Relaxed	Amazed	Awe	Delighted	Engaged	Enthusiastic	Excited
Happy	Inspired	Passionate	Angry	Annoyed	Frustrated	Proud
Accepting	Caring	Empathy	Safe	Fascinated	Interested	Sad
Despair	Disappointed	Heartbroken	Hopeless	Lonely	Confused	
Empty	Indifferent	Embarrassed	Shame	Ashamed	Afraid	
Frightened	Hesitant	Nervous	Worried	Helpless	Appreciative	
Delighted	Fortunate	Guilt	Regret	Hopeful	Encouraged	

Carbon Footprint

Your carbon footprint is the amount of greenhouse gases you produce. These gases are produced by things such as using electricity, driving cars and eating food that has been grown hundreds of miles away.

Write down ways you can lower your carbon footprint.

Turn off lights and appliances; walk, cycle or use public transport; eat less meat and more local food; recycle and reuse; use less heating and air conditioning; use renewable energy and plant trees.



9 OP 5a Rising Sea Levels Mind Map

1. Your teacher will show you the following videos to introduce the impact of rising sea levels on the earth.

[Rising Sea Levels](#)

[Greenhouse Gases](#)

[Rising Sea Levels from Space](#)

2. In **groups** you must address the following:

- Link between an increase in temperature and sea levels
- Urban impact from rising sea levels
- Issues with salt water mixing with freshwater
- Ice melting and sea levels
- Greenhouse gases

· Temperature and sea level: as global temperature rises, sea water expands (thermal expansion) and land ice melts, so sea level rises.

3. Using A3 paper provided by your teacher, create a **‘Mindmap’** of your group's ideas and elect a group leader to share your ideas from your mind map with all other students in your class.

· Salt water mixing with fresh water: sea water pushes into rivers and underground aquifers, making drinking and irrigation water too salty and killing freshwater plants and animals.

· Ice melting and sea level: melting land ice in Antarctica and Greenland adds water to the ocean and raises sea level, but melting floating sea ice does not.

· Greenhouse gases: burning fossil fuels releases CO₂, methane and nitrous oxide, which trap heat and cause the warming that melts ice and expands the oceans.

9 OP 5b Modelling Melting of Polar Ice Caps

Aim:

Have you ever noticed that if you leave an ice cube out on the kitchen counter and come back to check on it in a while, you find a puddle?

The same thing happens to ice in nature — if the temperature gets warm enough, the ice melts. In this science activity, you will explore what happens to sea levels if the ice at the North Pole melts, or if the ice at the South Pole melts.

Does melting ice at either cap contribute to a rise in sea levels? It is an especially important question for the millions of people who enjoy living along the coasts of the world.

Procedure

1. Put one cup of Play-Doh or modelling clay into one of the clear containers. This container will be a model of the South Pole, with the mound representing the continent of Antarctica. Use a butter knife to scrape around the sides of the measuring cup, if necessary. Make the top of the dough flat and level. Leave some space between the sides of the dough and the wall of the container all around, so that you can add water later.
2. Take your model of the South Pole and carefully add around $\frac{1}{4}$ cup of water around the sides of the dough, so that the water level comes up about one-third to half of the way up the dough mound. The water represents the ocean.
3. Place two ice cubes on top of the dough, lightly pressing them down into the dough. Immediately mark the water level on the side of the container with the permanent marker or colored tape. The ice cubes represent the southern polar ice sheet in this model of the South Pole.



What do you think will happen as the ice cubes melt?

The ice cubes will melt and the meltwater will run off the dough into the water around it, so the water level in the container will rise; this models what happens when ice on land (Antarctica) melts.

4. Take the second, clear container and fill it about one-third to half full of tap water. The water represents the ocean. Add two ice cubes to the container and immediately mark the water level on the side of the container with colored tape or a permanent marker. This container is a model of the North Pole, where the ice cubes represent the floating northern polar ice cap.



What do you think will happen as the ice cubes melt?

The ice cubes will melt and turn to water, but the water level should stay the same, because floating ice already displaces its own mass of water.

5. Allow the ice in your models to melt in a place where they will not be disturbed. Keep an eye on the ice cubes in the South Pole model to make sure they stay balanced on the dough, and that the water from these melted ice cubes is able to drain off of the dough. Once the ice has completely melted, check the water level in each container again.



Has the water level risen in any of the containers? If it has, why do you think this is and what do you think the implications are for changes in sea level in the real world?

The water level rose only in the South Pole model, where the ice rested on land; it stayed the same in the North Pole model, where the ice was floating.

Conclusion:

From your observations draw a scientific conclusion that links your laboratory investigation to the real world issue of polar ice caps melting. Be as scientific as you can in your discussion !

Melting ice resting on land runs into the sea and raises sea level, while melting floating sea ice does not, so warming that melts land-based ice sheets and glaciers raises global sea level.

9 OP 5c Polar Ice Caps - Extension

Analysing Sea Level Rise

Use the satellite data graphs at [Sea Level | Vital Signs – Climate Change](#) to answer the questions below. (Note: To obtain exact data points, place your mouse on the section of the graph you would like to examine.)

1. What is the source of the data for each graph?

Both graphs use NASA data - satellite altimetry measurements of sea height and the longer tide-gauge record, as published on the NASA Sea Level / Vital Signs website.

2. Which years are covered by each graph?

The tide-gauge graph covers about 1880 to the present, and the satellite graph covers 1993 to the present.

3. Is one graph a better representation of global sea levels than the other?

Why or why not?

The satellite graph represents global sea level better today because it measures the whole ocean, while tide gauges only record a few coastal sites; but the tide-gauge record is longer.

4. By approximately how many millimetres did sea level rise between:

a) 1910 and 1930?

About 30 mm (roughly 1.5 mm per year).

b) 1930 and 1950?

About 40 mm (roughly 2 mm per year).

5. What is the approximate average rate of increase of sea level rise between 1900 and 2000?

About 1.7 mm per year on average between 1900 and 2000.

6. By how many millimetres did sea level rise between the first measurement obtained in January 1993 and the first measurement obtained in January 2013?

About 65 mm - from about 0 mm in January 1993 to about +65 mm in January 2013.

7. What is the approximate rate of sea level rise between January 1993 and present?

About 3.4 mm per year - roughly double the 20th-century average rate.

Examine the line graph for average global temperature at <http://climate.nasa.gov/vitalsigns/global-temperature> and use it to answer the questions below.

8. By how much did the average global temperature change, and did it increase or decrease between 1910 and 1930?

Between 1910 and 1930 the average global temperature increased by about 0.3 degrees C.

How about between 1930 and 1950?

Between 1930 and 1950 it changed very little - a slight decrease of about 0.1 degrees C.

9. Compare your answers to question number 8 with your answers to question number 4. Can you offer an explanation for the correlation or lack thereof?

1910-1930 matches, because warming expands sea water and melts ice so both rise. 1930-1950 does not: sea level kept rising while temperature levelled off, because the ocean responds slowly to warming.

10. What is the approximate average global temperature rise per year from the first measurement taken in 1880 to present?

About 0.01 degrees C per year (a total rise of roughly 1.2 degrees C since 1880).

9 OP 6a Modelling Oil Spills and Clean Up of Spills

Aim:

Essential question: What are the effects of both the spill and the cleanup? What can we, as humans, do to avoid these problems in the future?

In this practical activity you will create a model habitat, complete with living organisms, which will suffer an oil spill. They will then test different tools to try to clean their habitat as best as possible.

Procedure:

1. Using the equipment that has been provided, you must use the rocks to form an island that rises out of the water in the tub.
2. Next, add “plants” (sticks and lettuce or moss) and “animals” (feathers). Some can be on the land, but most should be in the water. Make sure you are taking turns building the island so everyone participates!
3. Next pour the oil into the tub, spreading over the island, water and habitat items.

Observe and discuss: How is each material reacting with the oil? Record their observations in the results table below.

· Observations: the oil floats and spreads as a slick; it coats the rocks, sticks, lettuce and feathers. Feathers clump, become heavy and lose waterproofing; plants become coated and droop.

4. Use each cleaning tool to remove, or barricade, oil from the rocks, water and habitat items. Make notes about how each tool works in your results table below.
5. After testing each tool, discuss your observations about the tool as a group.

Results:

Use this table to record how the habitat materials interact with the oil.						
Material	Water	Rock	Feather	Stick	Fur	Plants
Reaction with Oil						

Use this table to evaluate each of the cleaning tools.				
Cleaning Tool	Pros	Cons	What habitat material does it clean the best?	What effect does it have on the environment?
String				
Sponge				
Cotton Ball				
Spoon				
Dish Soap				
Eyedropper				

· Results (model): oil floats as a slick; it sticks to the rocks and coats the sticks and lettuce; feathers clump and lose waterproofing; oil is very hard to remove from feathers and moss.

Conclusion and Analysis:

1. What happened to the land and habitat items when the oil spilled?

The oil spread over the water and the land, coating the rocks, plants and feathers; the feathers clumped, became heavy and were no longer waterproof, and the plants were smothered.

2. How did your group use the different tools? Why?

We used each tool in turn - a boom to contain the oil, absorbent pads to soak oil off the water and spoons or pipettes to lift oil from the rocks - so we could see which tool removed the most oil.

3. Which tools worked best for cleanup? Why?

The absorbent pads and the skimmer worked best, because they soak up or lift the oil off the water surface and can then be taken away, whereas other methods only spread the oil around.

4. What was the hardest material in your habitat to get the oil off of? Why do you think this is?

The feathers were hardest, because oil soaks into their fine, textured surface and sticks to them; the smooth rocks could be wiped clean more easily, and the moss/lettuce was also difficult.

5. What does this tell us about real world oil spills? What are some other ways we could clean up oil spills?

It shows real spills are very hard to clean up completely - oil spreads fast, sticks to wildlife and can kill it.
Other methods: dispersants, booms and skimmers, burning the oil, bioremediation.

6. Is there anything else we can do to minimise the impact of oil spills?

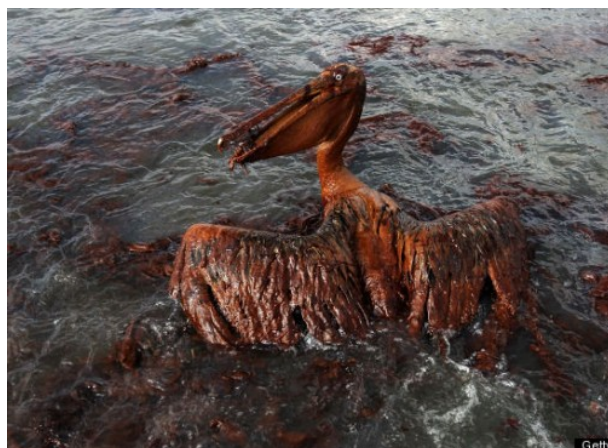
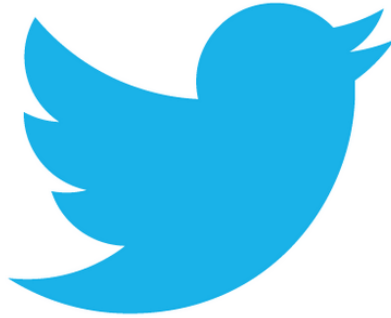
Yes - prevent spills first: double-hulled tankers, better maintenance and inspection, stricter laws, fast response teams, and less oil use by switching to renewable energy.

9 OP 6b Oil Spills - Tweet to Raise Awareness

Imagine the oil spill has just happened! Write a tweet to raise awareness of the issue.....

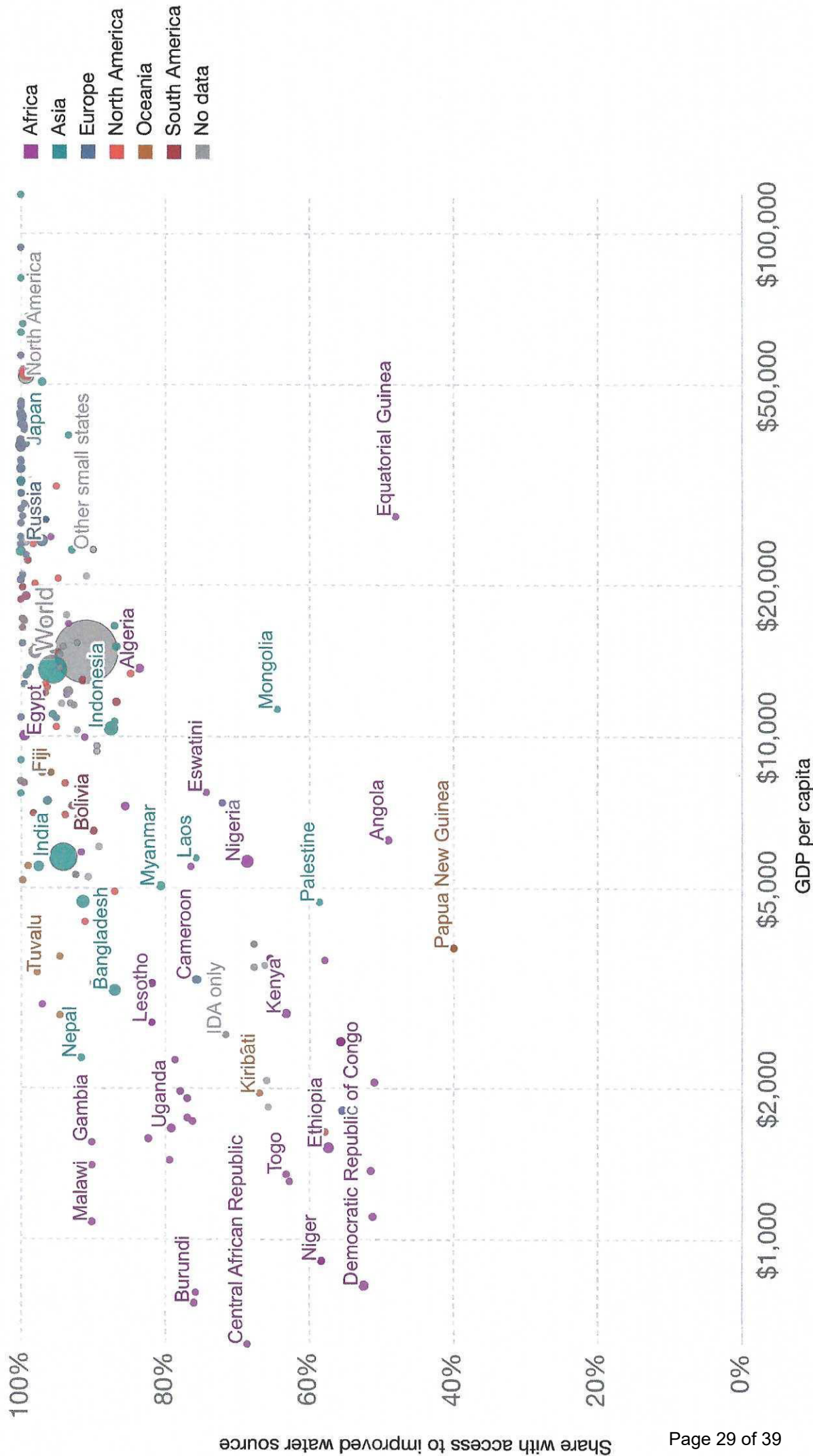
Use less than 140 characters.

Oil is choking our ocean. Birds, fish and reefs are dying. We need safer shipping and clean energy now. #OilSpill
#OurPlanet



Improved water sources vs. GDP per capita, 2015

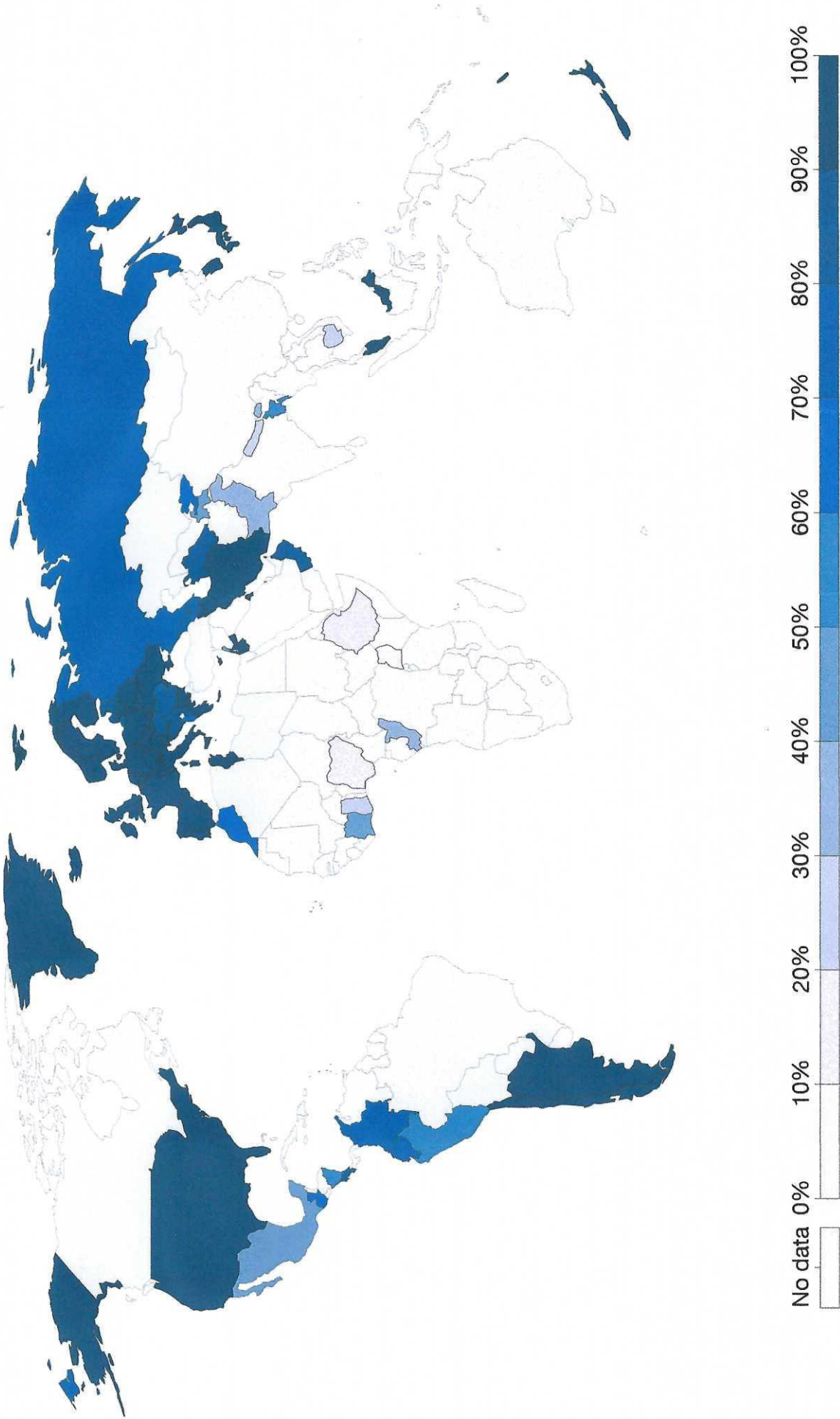
An improved drinking water source includes piped water on premises and other sources (public taps or standpipes, tube wells or boreholes, protected dug wells, protected springs, and rainwater collection). GDP per capita is measured in constant 2011 international- $\text{\$}$.



Share with access to improved water source

Share of the population using safely managed drinking water, 2015

A safely managed drinking water service is defined as one located on premises, available when needed and free from contamination.



Filtering Contaminated Water

Introduction:

In this experiment you will filter three water samples with different "contaminant" (food colour) concentrations using three different filter types

- 1) powdered activated carbon
- 2) granular activated carbon
- 3) no carbon filter

Once you have completed the filtration, you will assess the effectiveness of each method by looking at the colour change of your samples. You are given 3 samples of water:

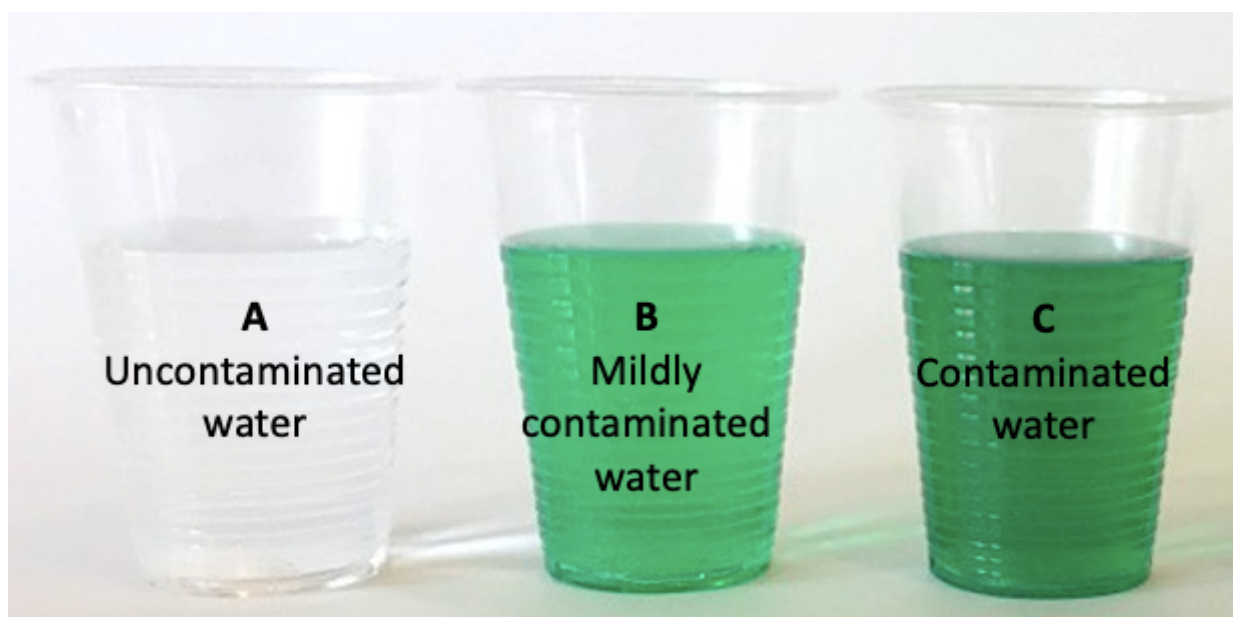


Image showing three samples with different concentrations of "contaminant" (green food colouring)

Method:

1. Label three 250 cm³ beakers 1, 2 and 3.
2. Place 3 coffee filters over each beaker and secure them with an elastic band.
3. Measure 3 g of powdered activated carbon in a weighing boat on an electronic balance and pour the powder into the filter paper on beaker 1.
4. Measure 3 g of granulated activated carbon in a weighing boat on an electronic balance and pour the granules into the filter paper on beaker 2.
5. The filter paper on beaker 3 should not have anything added.
6. Use a measuring cylinder to measure 100 cm³ of solution A and pour it into beakers 1, 2 and 3.
7. Allow the solution to filter and write your observations of the filtrate in the results table.
8. Repeat each step with solution B and C.

Results table:

Write your observations in the table below. Use terms such as: no change, slightly paler, almost colourless, colourless.

Filter Type	Observations		
	A uncontaminated	B mildly contaminated	C very contaminated
Powdered carbon	no change (already clear)	almost colourless	slightly paler
Granulated carbon	no change	slightly paler	no change
No carbon	no change	no change	no change

Analysis and Conclusion:

Which filter type was best at removing the food dye?

The powdered activated carbon filter was the best at removing the food dye.

Explain why.

Powdered carbon has a much larger surface area than the granules, so more dye particles are adsorbed onto its surface, so more colour is removed from the water.

Investigating Wind Turbines

Results, analysis and conclusions

Angle of turbine blade (Degrees)	Potential difference (Volts)
Flat	0.0 V
10	0.6 V
20	1.4 V
30	2.2 V
40	1.5 V
50	0.7 V

Now draw a line graph to show the output:

X axis is what was changed (angle of turbine blade – independent variable)

Y axis is what was measured (potential difference produced – dependant variable)

Model data: potential difference rises from 0 V with flat blades to a maximum at 30 degrees, then falls as the angle increases to 50 degrees. Students' own readings will differ.

Which angle gave the highest potential difference?

30 degrees

It gave the largest potential difference, about 2.2 V.

How does the angle alter the speed at which the turbine rotates (indicated by the potential difference)?

With the blades flat the moving air pushes them edge-on, so the turbine hardly turns and the potential difference is 0 V.

As the angle increases to about 30 degrees the blades catch more of the moving air, so the turbine spins faster and the potential difference rises.

Above 30 degrees the blades are at too steep an angle, so the air flows past them instead of pushing them and the turbine slows down.

So the potential difference falls again - the blades work best at about 30 degrees, giving the fastest rotation and the largest voltage.

Now I have found out which is the best angle to put the blades at, other variables I could investigate include:

the number of blades on the turbine

the length (or area) of the blades

the shape or material of the blades

the wind speed, by moving the fan nearer or further away



MYP Science Practical Report Template

Names of Everyone in Group

Form:

UNIT:	Our Planet
Investigation Title:	Investigating how the angle of the turbine blades affects the potential difference produced by a wind turbine.

Criterion B - Inquiring and Designing

Research Question

(What are you trying to find out?)

How does changing the angle of the turbine blades (flat, 10, 20, 30, 40 and 50 degrees) affect the potential difference in volts produced by the turbine?

Background Scientific Information and Hypothesis

Moving air pushes the blades and turns the generator, which produces a potential difference. Flat or very steep blades catch less air, so the turbine spins more slowly and less voltage is made.

(What do you think will happen and why?)

I predict the largest potential difference at about 30 degrees, because at that angle the blades catch the most moving air, so the turbine spins fastest and the generator makes the biggest voltage.

Variables

Independent <i>(What will change and how will you change it?)</i>	The angle of the turbine blades. I will change it by setting the blades at flat, 10, 20, 30, 40 and 50 degrees with a protractor and repeat each angle.
Dependent <i>(What will you measure to gather results?)</i>	The potential difference in volts produced by the turbine, measured with a voltmeter connected across the generator. I will take three readings at each angle.
Control <i>(What variables will you keep the same to make it a fair test? How will you keep the variables constant?)</i>	Same fan speed and distance to the turbine, same number, size and material of blades, same generator and voltmeter, same room. I will use a clamp stand and metre rule to keep each one the same.

Apparatus Set-up

(Draw a labeled diagram of the assembled apparatus that you will use. Write the name of any equipment not shown in the diagram)

· Diagram: electric fan a fixed distance from the turbine blowing at the blades; blades set with a protractor; turbine clamped to a stand; generator wired to a voltmeter. Also needed: metre rule.

Method

*(Write a step by step account of how you conduct your experiment – the method must be DETAILED).
Add as many steps as you need to.*

- 1.
 - 2.
 - 3.
 - 4.
 - 5.
1. Clamp the turbine to a stand, connect the generator to the voltmeter and place the fan 30 cm from the blades.
 2. Set the blades at the first angle (flat) with the protractor and switch on the fan.
 3. Wait for the turbine to spin steadily, read the potential difference from the voltmeter and record it in a table.
 4. Repeat the reading twice more at the same angle and work out a mean.
 5. Repeat steps 2 to 4 for 10, 20, 30, 40 and 50 degrees, keeping the fan position and speed the same each time.

Safety

Keep hands, hair and loose clothing away from the spinning blades. Do not touch the blades while the fan is on, clamp the turbine so it cannot fall, and keep liquids away from the wires.
(How will you make sure your method is safe?)

Reliability

Take three readings at each angle and use the mean, repeat the whole experiment, keep all other variables the same (fan speed and distance) and check or repeat any anomalous reading.
(What will you do to ensure your results are reliable?)

Criterion C - Processing and Evaluating
Results

Raw Data

Raw data table: angle of the blades in degrees with three repeat readings of potential difference in volts and a mean; record your own measurements here.

Angle of blades (degrees)	Reading 1 (V)	Reading 2 (V)	Reading 3 (V)
Flat	0.0	0.0	0.0
10	0.6	0.7	0.6
20	1.4	1.3	1.4
30	2.2	2.1	2.2
40	1.5	1.6	1.5
50	0.7	0.7	0.8

Observations

(What changes did you notice as the experiment was conducted, e.g. colour change, bubbles, heat change?)

As the angle increased the turbine spun faster and the reading rose, peaking at 30 degrees. At 40 and 50 degrees the blades whooshed loudly and slowed, so the reading fell. No heat change was seen.

Graphing Results

You should draw a graph of your results and attach it to this report.

Analysis of Graph

Graph rises from 0 V with flat blades, peaks at 30 degrees (2.2 V), then falls to 0.7 V at 50 degrees. Voltage rises with angle up to 30 degrees then falls. The 0.8 V at 50 degrees may be an anomaly.
(What did your results show? Are there any trends or patterns in your results? Can you give an example of a pattern or trend from your results? Are there any anomalies in your results?)

Conclusion and Evaluation

Scientific Conclusion and Explanation of Results

Results support my hypothesis: the biggest voltage (2.2 V) was at 30 degrees, where the blades catch the most air and spin the generator fastest. Flat or steep blades let air pass, wasting energy.
(Summarize what your results show. How well does this support your hypothesis? Can you explain your results using scientific knowledge?)

Evaluation of Scientific Method and Results

(How reliable were your results? How well did your investigation answer the research question? What went well with your method?)

Results were fairly reliable as each angle was repeated three times with close readings. The pattern was clear and the method worked well because the fan speed, distance and blades stayed the same.

(Were there any problems or parts of your investigation that were not very accurate or could have been done better?)

Issue with justification	Suggested SCIENTIFIC Improvement
	Issue: the fan speed may have varied during the test. Improvement: use a fan with a fixed speed setting and check the wind speed with an anemometer before each reading.
	Issue: the blade angle was only estimated with a protractor. Improvement: fix a printed protractor scale behind the blades so each angle is set exactly, and test 25 and 35 degrees as well.
	Issue: only three repeats were taken, so one anomaly affects the mean. Improvement: take five readings at each angle, calculate a mean and repeat any anomalous value.
	Issue: the distance to the fan was not measured exactly. Improvement: use a metre rule to keep the fan exactly 30 cm away, and use the same blades and voltmeter for every test.

Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p10: Blank line graph: plot each year (1970-2019) against the forest area and join the points - the line falls steadily from 4.10 million km² in 1970 to 3.38 million km² in 2019.

p20: Urban impact: coastal cities flood, roads, homes and sewers are damaged, beaches and farmland are lost, drinking water becomes salty, sea defences cost billions and people are forced to move.